Presentation 3 - Celena Kim

Presentation Link:

- [Power BI Dashboard]
- [Callback & Post-Queue Pickup Analysis Code]
- [Presentation 3 Slides]

What did you do:

- For this presentation, I joined Jake and Logan in analyzing the CAR dataset. Building off of Jake and Logan's analyses, which mainly focused on queue wait times and callbacks vs. no callbacks, I wanted to focus on analyzing what happens after a customer connects with an agent. Specifically, I examined the duration of calls for individuals who used the callback feature versus those who stayed in the queue to wait for the agent. I defined a session as utilizing the callback feature if it contained at least one activity where flow_name == "CourtesyCallback". For these sessions, I determined whether the callback was successful if they eventually reached an agent (indicated by the last instance of "LegalServerScreenPop" in the activity_name column) and if the rows following this point did not immediately contain a termination reason such as "Customer Left", "Agent Left", "No Answer From Agent", etc. I then calculated the call duration from the start time of the last LegalServerScreenPop activity to the last row in that call session. For the post-queue pickup analysis, I identified sessions that did not request callbacks but included queue-related activities such as PreQueue messages or Queue menus. In a similar fashion, call duration was calculated as, following a queue, the time elapsed between when the session reached an agent (the last LegalServerScreenPop timestamp) and the end of the session. After calculating the call durations for these two call types, I then visualized two graphs within Power BI: the overall number of calls per hour by call type, and the average duration of calls per hour by call type. I also included cards displaying the average duration of each call type, as well as a date slicer to filter by different periods.

How does it help the project?:

- This analysis provides valuable insights into comparing how calls are handled as a callback vs. post-queue pickups. The results show that callbacks not only are significantly more prevalent but also slightly longer in duration, suggesting that individuals strongly prefer to utilize the callback feature rather than wait on hold in a queue. Moreover, agents may either be spending more focused time with individuals during callbacks, leading to the slightly longer average durations, or should streamline their processes in order to decrease these call durations. These insights can inform call optimization by prioritizing resources and training towards handling these callbacks more efficiently since they represent the majority of customer interactions. Furthermore, visualizing these trends by the hour reveals that peak call times for both callback and post-queue pickups are 8 AM, 12 PM, and 4 PM, suggesting that adequate staffing and workflow adjustments may be necessary to manage these high-traffic periods effectively. Overall, this insight into agent call handling highlights clear patterns and opportunities to improve caller experience by focusing on timing, prevalence, and management of callbacks vs. post-queue calls.

Issues faced (if any):

- One issue I faced was not being too sure if my logic for defining successful callbacks/post-queue pickups and calculating call durations was correct. There were some distinct cases such as a call session containing multiple instances of "LegalServerScreenPop" that I was unsure how to interpret.
- For instance, contact session id "01a50ccf-91f5-403c-92ae-1b77bba1dff4" had three instances of "LegalServerScreenPop" across two days, with two different agents, Ana Franco and Marisol Guadarrama. I was unsure whether all three of these callback instances should have been considered as 3 distinct callback sessions.

Attempts to resolve issues

- To resolve this uncertainty, I set a consistent approach to just consider the last instance of "LegalServerScreenPop", coupled with the absence of a termination reason, to represent a successful callback or post-queue pick up. I also calculated call duration as the point of the last instance of "LegalServerScreenPop" to the last row of the call session.

Next Steps

- I plan to revisit my logic for defining callback/post-queue pickup success by attempting to handle multiple callback instances such as what we saw with contact session id "01a50ccf-91f5-403c-92ae-1b77bba1dff4". Once this logic is clarified, I would like to focus in on this analysis in more detail, such as adding in a filter for different queue types.

Presentation 2 - Celena Kim

Presentation Link:

- [Power BI Dashboard]
- [Call Duration Data Prep Code]
- [Presentation 2 Slides]

What did you do:

- For this analysis, I used the All Calls dataset to visualize call counts and durations for distinct called numbers. I first defined Legal Aid's open and closed hours and created a reference table to map phone number descriptions from the "Popular Numbers (AllCallsData)" spreadsheet. I then removed NA called numbers and any duplicate rows that corresponded to the same call log, which reduced the dataset from 1,001,537 rows to 732,420 rows. Next, to enable an interactive filtering feature in the dashboard, I created several additional columns: call path (with values inbound, outbound, and internal, derived using the call direction and PSTN name), date, weekday vs. weekend, and open vs. closed hours. This prepared dataset allowed me to graph the distribution of call count and duration and categorize the calls by multiple dimensions of filters, enabling deeper insights into call patterns at both working and non-working hours.

How does it help the project?:

- By exploring call counts and durations across numerous categories of call path, day of the weekend, and open vs. closed hours, this analysis helps the project by identifying any trends or bottlenecks in service demand, highlighting peak periods or unnecessarily long call durations at various points, and enabling the Legal Aid to make any decisions to improve call response times and efficiencies. Specifically, the dashboard revealed that during closed hours on weekdays, the Farmworker main number (Migrant Legal Assistance Program) line displayed significantly high average durations of 1,182 minutes, suggesting a need to evaluate the process implemented for handling calls outside working hours and potentially implement some sort of automated system that immediately informs callers that the office is closed and points them to online resources, rather than letting those calls linger on the line. Additionally, on weekends when the office is completely closed, both the Farmworker main number as well as the "Transfer to the English main menu" lines exhibited unusually high average durations of around 1,282 minutes and 1,859 minutes, respectively, indicating that there may be a potential bottleneck within these lines on weekends with callers either waiting excessively or somehow reaching the unstaffed line. For this potential issue, a revised automated messaging process could also reduce this line’s high duration and provide alternative resources to these weekend callers.

Issues faced (if any):

- One issue I faced is not being too sure that I prepped my data correctly. Based on the feedback from project 1, rather than grouping by call direction (Terminating/originating), I created a column called "Call Path" with values inbound, outbound, or internal. This was derived by using the call direction and PSTN name, and utilized within the dashboard as a filter. Next, in order to not overcount duplicate logs of the same call, I removed any duplicate rows corresponding to the same combination of `Correlation ID` and `Called number`, and same combination of `Correlation ID`, `Start time`, and Duration. However, this reduced the dataset from 1,001,537 rows to 732,420 rows, meaning some analyses were limited to smaller subsets. Given that my last recommendation from the previous presentation was to avoid analyses that only focus on a small subset of calls, I attempted to produce results that focused on higher proportions of the dataset, however, I still found it difficult to extract meaningful insights from larger subsets, since clear bottlenecks were found in areas that represented 11% of the dataset. Additionally, many of the phone numbers did not have descriptions, so it was difficult to interpret some of my results.

Attempts to resolve issues/Next Steps

- In order to address producing insights for larger portions of the dataset, plan to analyze more significant categories for analysis. Specifically, since I was able to identify that the main line called number category clearly holds the largest proportion of calls from this analysis, I could focus on this subset and explore the different pathways that extend from the main line to identify any potential bottlenecks. For instance, perhaps a pathway for a caller to get to a certain line after being funneled through the main line is exorbitantly long-winded and tedious. By zooming in on a known significant portion of the dataset, I am more likely to uncover actionable patterns that can reflect the broader call landscape, rather than pinpointing minor improvements.

Presentation 1 - Celena Kim

Presentation Link:

- Power BI Dashboard
- No extra code files– DAX measures and new columns were created within Power BI:
    - Avg Duration = AVERAGE(all_calls[**Duration**])
    - Call_Column = 'all_calls'[**Direction**] & " | " & 'all_calls'[**Releasing party**] & " | " & 'all_calls'[**Call outcome**]

What did you do:

- To identify potential inefficiencies within Legal Aid's call process, I analyzed patterns in average call duration across various call combinations. Using the All Calls dataset, I created a DAX measure in Power BI to calculate the average durations for each unique combination of Direction (Terminating in Legal Aid or Originating), Releasing Party (Local, Remote, or Unknown), and Outcome (Refusal, Failure, or Success). I then visualized these results in a graph to display this distribution of durations. Finally, to make key patterns more visible, I applied conditional color formatting to highlight combinations with notably higher durations in red. Three distinct outliers emerged from this analysis, all with a common trait: having an "Unknown" releasing party.

How does it help the project?:

- This finding highlights a clear opportunity to optimize how the system manages calls with an "Unknown" releasing party. Since these calls tend to have significantly longer average durations, Legal Aid could improve their efficiency in various ways such as by automatically terminating unknown caller IDs, implementing a timeout mechanism for these unknown calls instead of letting them linger on the line, or further investigating why certain callers appear as unknown in order to allow these callers to still receive the help they deserve. These solutions would not only prevent wasted call times, but also allow for legitimate callers with these unknown caller IDs to still receive assistance.

Issues faced (if any):

- After displaying the counts of these 3 "Unknown" call combinations with the bar chart on the right side of the dashboard, I found that these instances are very few in number within the vast 1 million rows of calls in the dataset, with counts only ranging from 2-158. Therefore, while these outliers do highlight an interesting inefficiency, the overall impact of this issue is minimal in scope. This limitation suggests that while the dashboard does flag a bottleneck within the call process, further analysis is necessary to identify an inefficiency that has a larger impact on overall performance.

Attempts to resolve issues/Next Steps

- While I am still interested in this pattern of lingering call durations, I plan to identify more widespread patterns that have a greater impact within the scope of this large dataset. For instance, I could attempt to analyze different call combinations with higher frequencies to uncover duration inefficiencies, or explore other areas for optimization besides such as agent response times. Overall, I aim to gain a more comprehensive understanding of where the system could be improved by highlighting prolonged interactions or delays that may be reducing call efficiency, while keeping in mind to identify business insights that have a meaningful effect at scale.